

Richard Zhuang

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EDUCATION

- **Stanford University** Stanford, CA, U.S.A
M.S. in Computer Science 2025. 9 – 2027. 6
- **University of California, Berkeley** Berkeley, CA, U.S.A
B.A. Double Majored in Computer Science (Honor) and Applied Mathematics 2021. 9 – 2025. 5
 - **GPA: 4.0/4.0**
 - **Course Highlights:** Machine Learning, Deep Learning, Foundations of LLMs, Statistical Learning Theory, Probability and Stochastic Processes, Data Science Techniques, Design and Analysis of Algorithms.

SKILLS

General: Python, Java, C, JavaScript, Linux, Git **Data Science:** Pandas, Matplotlib, PostgreSQL, R
Machine Learning: PyTorch, Numpy, Transformers, vLLM, veRL, Llama-Factory, Accelerate, Huggingface

EXPERIENCES

- **Training Agentic Tool-Use LLM with Reinforcement Learning** Mountain View, CA
Post-Training Engineer at Bespoke Labs (bespokelabs.ai) Jan 2025 - May 2025
 - **Extended Bespoke’s open-source data-curation toolkit Curator (1.5 K GitHub stars):** Implemented per-question sampling-parameter support to enable fine-grained data filtering for large-scale post-training pipelines; authored working demo and merged 2 PRs into the main repository.
 - **Trained multi-turn tool-use agents with GRPO:** Improved Qwen2.5-7B-Instruct’s performance on BFCL by **23 % (55 % → 78 %)** using an RL recipe incorporating overlong-filtering + KL stabilization; summarized findings in a technical blog (**≥ 200K views on X**) that serves as a practitioner’s guide to stable multi-turn RL.
 - **Spearheaded internal RL training framework for data-science agents:** Designed a new rollout generator to orchestrate agent-environment interactions inside Daytona sandboxes, enabling 64 concurrent rollouts for scalable RL training.
- **OpenThoughts-Agent: Data Recipe for Training Agents** Stanford, U.S.A.
Researcher at Stanford advised by Prof. Ludwig Schmidt Sep 2025 - Present
 - **Developed automatic environment curation pipeline:** Created automated system using Mini-SWE-Agent that turns arbitrary GitHub repos into interactive debugging environments with mechanistic reward signals.
 - **Architected large-scale SQL database for result tracking:** Designed relational schema linking (estimated) **≥ 3,000** training and evaluation runs (~2–3 TB artifacts), enabling efficient leaderboard generation and aggregated performance analysis across models.
- **Advancing LLM Reasoning using Collective Intelligence** Menlo Park, U.S.A.
Researcher at Stanford advised by Prof. Christopher Ré Sep 2025 - Present
 - **Executed systematic trace analysis across teacher LLMs:** Analyzed datasets from QwQ-32B, DeepSeek-R1, and Claude-3.7-Sonnet to characterize distributional properties in reasoning data.
 - **Conducted 50+ Distillation experiments:** Performed Supervised Fine-tuning on Llama-3.2-3B to examine correlation between student performance and teacher properties such as verbosity and reasoning strategy diversity.

SELECTED PUBLICATIONS

- **EmbedLLM: Learning Compact Representations of LLMs** ICLR 2025 Spotlight
Richard Zhuang, Tianhao Wu, Zhaojin Wen, Andrew Li, Jiantao Jiao, Kannan Ramchandran
 - **High-Level Summary:** Proposed an encoder-decoder method that learns compact embeddings as “LLM identities” for **accurate model routing and performance forecasting**.
- **PokerBench: Training LLMs to Become Professional Poker Players** AAAI 2025
Richard Zhuang, Akshat Gupta, Richard Yang, Aniket Rahane, Zhengyu Li, Gopala Anumanchipalli
 - **High-Level Summary:** Introduced a benchmark assessing LLMs’ strategic reasoning; fine-tuned models **outperformed GPT-4** in decision quality in poker gameplay.